

Course Syllabus: CS401 - Artificial Intelligence & Deep Learning

Department of Computer Science & Engineering | Academic Year 2024-2025

Course Overview

This course provides a comprehensive introduction to Artificial Intelligence, Machine Learning techniques, and Deep Learning architectures. Students will gain practical and theoretical understanding of intelligent search, neural networks, and computer vision.

Unit 1: Introduction to AI & Search Algorithms

Foundations of AI, Problem Formulation, State Space Search, Uninformed Search strategies (BFS, DFS), Informed Search (Greedy Best-First, A* Algorithm), Heuristic Functions, Adversarial Search and Game Playing (Minimax Algorithm, Alpha-Beta Pruning).

Unit 2: Machine Learning Fundamentals

Supervised vs Unsupervised Learning, Linear Regression, Logistic Regression for Classification, Decision Trees, Random Forests, Support Vector Machines (SVM), Evaluation Metrics (Accuracy, Precision, Recall, F1-Score, ROC-AUC curve), Overfitting and Regularization Techniques (L1, L2).

Unit 3: Deep Learning & Computer Vision

Multilayer Perceptrons (MLP), Backpropagation, Activation Functions (ReLU, Sigmoid, Softmax), Convolutional Neural Networks (CNN), Convolutional and Pooling Layers, Image Classification, Object Detection (YOLO), Transfer Learning (ResNet, VGG).

Unit 4: Natural Language Processing (NLP)

Text Preprocessing, Tokenization, Stemming, Lemmatization, Bag of Words, TF-IDF, Word Embeddings (Word2Vec, GloVe), Recurrent Neural Networks (RNN), LSTM, Attention Mechanism, Transformer Architectures (BERT, GPT), Sentiment Analysis.

Unit 5: AI Ethics & Model Deployment

Bias and Fairness in Machine Learning models, AI Ethics and Safety, Explainable AI (SHAP, LIME), Model Compression (Quantization, Pruning), Model Deployment using FastAPI and Docker, Continuous Monitoring.

Evaluation & Grading Scheme

- Internal Assessment & Assignments: 20%
- Mid-Semester Examination: 20%
- Mini-Project Presentation: 20%
- End-Semester Final Examination: 40%